Data Visualization Techniques

Unit-I

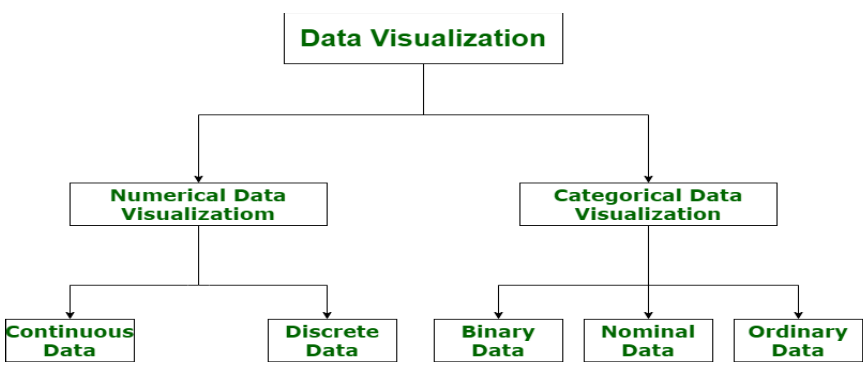
1.1 Introduction: Visualization as the communication of information using graphical representations. Pictures have been used as a mechanism for communication since before the formalization of written language. A single picture can contain a wealth of information, and can be processed much more quickly than a comparable page of words. This is because image interpretation is performed in parallel within the human perceptual system while the speed of text analysis is limited by the sequential process of reading. Pictures can also be independent of local language, as a graph or a map may be understood by a group of people with no common tongue.

Data visualization is a powerful tool for enhancing understanding and communication of complex data. It involves representing data in a graphical or pictorial form, making it easier to understand and interpret. The importance of data visualization in effectively communicating and analyzing data to provide insights into the various types of data visualization tools and techniques available. Data visualization tools can be broadly classified into three categories:

Spreadsheets, data visualization software, and programming libraries.

- **Spreadsheets** - such as Microsoft Excel and Google Sheets, are one of the most common data visualization tools used in various domains. They provide basic data visualization capabilities, such as bar charts, line graphs, and scatter plots.
- **Data Visualization Software** is a specialized tool designed for data visualization and analysis. Examples of data visualization software include Tableau, Qlik View, and Power BI. These tools provide advanced data visualization capabilities, including interactive dashboards, heat maps, and network diagrams.
- **Programming Libraries** such as Matplotlib, ggplot2, and D3.js, are a type of data visualization tool that can be used to create custom data visualizations.

Categories of Data Visualization: Data visualization is very critical to market research where both numerical and categorical data can be visualized that helps in an increase in impacts of insights and also helps in reducing risk of analysis paralysis. So, data visualization is categorized into following categories:



i. Numerical Data Numerical data is also known as quantitative data. Numerical data is any data where data generally represents amount such as height, weight, age of a person, etc.

Numerical data is categorized into two categories:

- Continuous Data – It can be narrowed or categorized (Example: Height measurements).
- Discrete Data **–** This type of data is not "continuous" (Example: Number of cars or children's a household has). Examples are Pie Charts, Bar Charts, Averages, Scorecards, etc.

ii. Categorical Data Categorical data is also known as qualitative data. Categorical data is any data where data generally represents groups. It simply consists of categorical variables that are used to represent characteristics such as a person's ranking, a person's gender etc.

- Binary Data **–** In this, classification is based on positioning (Example: Agrees or Disagrees).
- Nominal Data **–** In this, classification is based on attributes (Example: Male or Female).
- Ordinal Data **–** In this, classification is based on ordering of information (Example: Timeline or processes).

It is an interesting exercise to consider the number and types of data and information visualization that we encounter in our normal activities. Some of these might include

- a simple formatted number representing a single item of interest, such as the gross national product (GNP) of the U.S.
- a table in a newspaper, representing data being discussed in an article;
- a train and subway map with times used for determining train arrivals and departures
- a map of the region to help select a route to a new destination
- a weather chart showing the movement of a storm front that might influence your weekend activities
- a graph of stock market activities that might indicate an upswing (or downturn) in the economy
- a plot comparing the effectiveness of your pain killer to that of the leading brand
- a 3D reconstruction of your injured knee, as generated from a CT scan
- an instruction manual for putting together a bicycle with views specific to each part as it is added
- a highway sign indicating a curve, merging of lanes or an intersection.

Why Data Visualization Important: There are many reasons why visualization is important. Perhaps the most obvious reason is that we are visual beings who use sight as one of our key senses for information understanding. The two examples below highlight why visualization is so important in decision making and the role of human preferences and training. One example focuses on data distortion, and the other on human interpretation.

The importance of data visualization is simple it helps people see, interact with, and better understand data. Whether simple or complex, the right visualization can bring everyone on the same page, regardless of their level of expertise.

- **Analyzing the Data in a Better Way** Analyzing reports helps business stakeholders focus on the areas that require attention. The visual mediums help analysts understand the key points needed for their business.
- **Faster Decision Making** If the data communicates well, decision-makers can quickly take action based on the new data insights, accelerating decision-making and business growth simultaneously.
- **Making Sense of Complicated Data** visualization allows business users to gain insight into their vast amounts of data. It benefits them to recognize new patterns and errors in the data. Making sense of these patterns helps the users pay attention to areas that indicate red flags or progress.

Different types of visualizations: General Types of Visualizations:-

- **Chart:** Information presented in a tabular, graphical form with data displayed along two axes. Can be in the form of a graph, diagram, or map.
- **Table:** A set of figures displayed in rows and columns.
- Graph: A diagram of points, lines, segments, curves, or areas that represents certain variables in comparison to each other, usually along two axes at a right angle.
- **Geospatial:** A visualization that shows data in map form using different shapes and colors to show the relationship between pieces of data and specific locations.
- **Info graphic:** A combination of visuals and words that represent data. Usually uses charts or diagrams.
- **Dashboards:** A collection of visualizations and data displayed in one place to help with analyzing and presenting data.

History of Visualization:

Early Visualization Perhaps the first technique for graphically recording and presenting infor-mation was that used by early man. An example is the early Chauvet-Pont-d'Arc Cave, located near Vallon-Pont-d'Arc in southern France .The Chauvet Cave contains over 250 paintings, created approximately 30,000 years ago. These were likely meant to pass on information to future generations. The oldest writing systems used pictures to encode symbols and whole words. Such systems are called *logograms.* The Kish limestone tablet is considered the earliest written document. It is from Mesopotamia and is mostly pictographic, but it has the beginning of syllabic writing found in cuneiform scripts. It is located in the ashmolean Museum Oxford

Another early writing system which came from the ancient Egyptians is called *hieroglyphics*. Hieroglyphs are divided into three major categories: logograms, phonograms, and determinatives. Hieroglyphic logograms are signsth at represent morphemes "the smallest language unit that carries a semantic interpretation". Phonograms are signs that represent one or more sounds. Determinatives are signs that help to join logograms and photograms together to disambiguate the meaning of a sequence of glyphs. Early visualizations came about out of necessity for travel, commerce, religion, and communication. Maps provided support for travelers where planning or survival was the key.

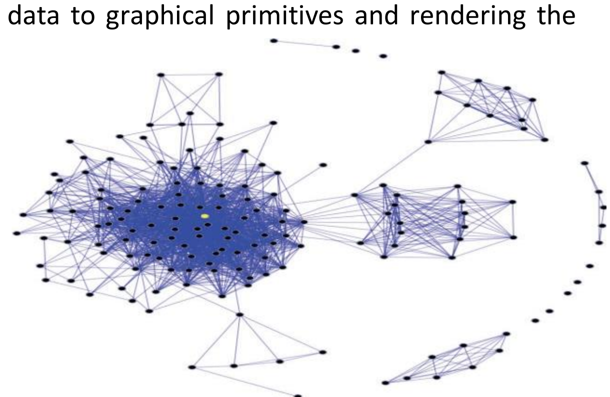
Visualization Today most often provides different levels of both qualitative and quantitative views of the information being communicated. a map of the Tokyo underground. It provides an easy-to-read yet logically distorted view of the criss-crossing network of the subway system to facilitate interpretation; similar techniques have been used for a number of subway and other transportation systems as well as for processes and their flows. Distorted views are often interpreted imprecisely and depend on the viewer. Most representations of two-dimension al maps exhibit some degree of distortion due to the 3D-to-2D projection (globe to plane). However, since a small area on the globe is well approximated by a plane, local street maps have minimal distortions and thus can provide strong relative detail and especially link information including connectedness closeness above,or below. Google map directions between two local street addresses, indicating the roads, intersections, and turns to make throughout the trip.

1.1.1 Relationship between Visualization and other Fields

i. Difference between Visualization and Computer Graphics:

Visualization was considered a subfield of computer graphics, primarily because visualization uses graphics to display information via images. As illustrated by any of the computer-generated images shown earlier, visualization applies graphical techniques to generate visual displays of data. Here, graphics is used as the communication medium.

In all visualizations, one can clearly see the use of the graphics primitives (points, lines, areas, and volumes). Beyond the use of graphics, the most important aspect of all visualizations is their connection to data. Computer graphics focuses primarily on graphical objects and the organization of graphic primitives; visualizations go one step further and are based on the underlying data and may include spatial positions populations or physical measures. Consequently visualization is the application of graphics to display data by mapping data to graphical primitives and rendering the display.



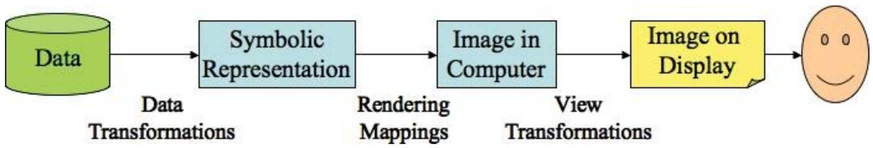
However, visualization is more than simply computer graphics. The field of visualization encompasses aspects from numerous other disciplines, including human-computer interaction, perceptual psychology, databases, statistics, and data mining, to name a few. While computer graphics can be used to define and generate the displays that are used to communicate the information, the sources of data and the way users interact and perceive the data are all important components to understand when presenting information.

Our view is that computer graphics is predominantly focused on the creation of interactive synthetic images and animations of three-dimensional objects, most often where visual realism is one of the primary goals. A secondary application of computer graphics is in art and entertainment, with video games, cartoons, advertisements, and movie special effects as typical examples. Visualization, on the other hand, does not emphasize visual realism as much as the effective communication of information. Many types of visualizations do not deal with physical objects, and those that do are often communicating attributes of the objects that would normally not be visible, such as material stress or fluid flow patterns. Thus, while computer graphics and visualization share many concepts, tools, and techniques, the underlying models (the information to be visualized) and goals (what the viewer hopes to extract) are fundamentally different.

ii. Scientific Data Visualization vs. Information Visualization

Although during the 1990s and early 2000s the visualization community differentiated between scientific visualization and information visualization, we do not. Both provide representations of data. However the data sets are most often different. Figure 1.30 highlights the importance and value of having both. Bio molecular chemistry, which once only considered the visual representation of molecules as stick and balls, has migrated over time to representations as spheres with rods, to more realistic ones. The view that both scientific visualization and information visualization are allied fields. In some cases, the data being visualized begs for different handling: a large volume (1K × 1K × 1K = 1 billion points) requires dealing with large numbers of memory accesses and large-scale computations, whereas displaying a scatter plot of a million patients from a file is more concerned with reading the data from the database or file; the computations are much simpler.

1.1.2 The Visualization Process: The designer of a new visualization most often begins with an analysis of the type of data available for display and of the type of information the viewer hopes to extract from or convey with the display. The data can come from a wide variety of sources and may be simple or complex in structure. The viewer may wish to use the visualization for exploration (looking for "interesting" things), to confirm a hypothesis (either conjectured or the result of quantitative analysis), or to present the results of one's analysis to an audience. Examples of interesting results are anomalies (data that does not behave consistent with expectations), clusters (data that has sufficiently similar behavior that may indicate the presence of a particular phenomenon), or trends (data that is changing in a manner that can be characterized, and thus used for predictive models).

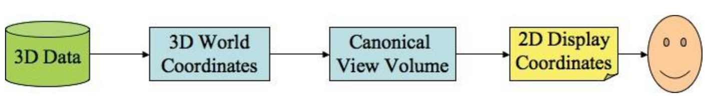


To visualize data, one needs to define a mapping from the data to the display. There are many ways to achieve this mapping. The user interface consists of components, some of which deal with data needing to be entered, presented, monitored, analyzed, and computed. These user interface components are often input via dialog boxes, but they could be visual representations of the data to facilitate the selections required by the user. Visualizations can provide mechanisms for translating data and tasks into more visual and intuitive formats for users to perform their tasks.

Another significant, yet often overlooked, component of the visualization process is the provision of interactive controls for the viewing and mapping of variables (attributes or parameters). While early visualizations were static objects, printed on paper or other fixed media, modern visualization is a very dynamic process, with the user controlling virtually all stages of the procedure, from data selection and mapping control to color manipulation and view refinement. There is no formula for guaranteeing the effectiveness of a given visualization. Different users, with different backgrounds, perceptual abilities, and preferences, will have differing opinions on each visualization.

i. The Computer Graphics Pipeline

- **Modeling**. A three-dimensional model, consisting of planar polygons defined by vertices and surface properties, is generated using a world coordinate system.
- **Viewing**. A virtual camera is defined at a location in world coordinates, along with a direction and orientation (generally given as vectors). All vertices are transformed into a viewing coordinate system based on the camera parameters.
- **Clipping**. By specifying the bounds of the desired image (usually given by corner positions on a plane of projection placed in front of the camera), objects out of view can be removed, and those that are partially visible can be clipped. Objects may be transformed into normalized viewing coordinates to simplify the clipping process. Clipping can actually be performed at many different stages of the pipeline.



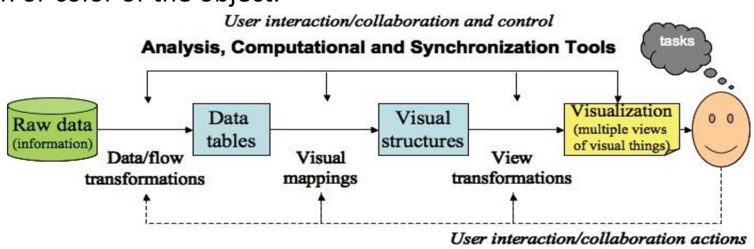
- **Hidden surface removal** Polygons facing away from the camera, or those obscured by others, are removed or clipped. This process may be integrated into the projection process.
- **Projection** Three-dimensional polygons are projected onto the two dimensional plane of projection, usually using a perspective transformation. The results may be in a normalized 2D coordinate system or device/screen coordinates.
- **Rendering** The actual color of the pixels associated with a visible polygon depends on a number of factors, including the material properties being synthesized (base color, texture, surface roughness, shininess), the

type(s), location(s), color and intensity of the light source(s) the degree of occlusion from direct light exposure, and the amount and color of light being reflected off of other objects onto the polygon.

ii. The Visualization Pipeline

The data/information visualization pipeline has some similarities to the graphics pipeline, at least on an abstract level. The stages of this pipeline are as follows:

- **Data modeling** to be visualized whether from a file or a database, has to be structured to facilitate its visualization. The name, type, range, and semantics of each attribute or field of a data record must be available in a format that ensures rapid access and easy modification.
- **Data selection** Similar to clipping data selection involves identifying the subset of the data that will be potentially visualized. This can occur totally under user control or via algorithmic methods such as cycling through time slices or automatically detecting features of potential interest to the user.
- **Data to visual mappings** of the visualization pipeline is performing the mapping of data values to graphical entities or their attributes. Thus, one component of a data record may map to the size of an object, while others might control the position or color of the object.

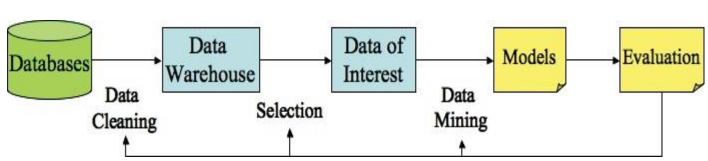


- **Scene parameter setting** As in traditional graphics the user must specify several attributes of the visualization that are relatively independent of the data. These include color map selection sound map and lighting specifications (for 3D visualizations).
- **Rendering or generation of the visualization**. The specific projection or rendering of the visualization objects varies according to the mapping being used techniques such as shading or texture mapping might be involved, although many visualization techniques only require drawing lines and uniformly shaded polygons.

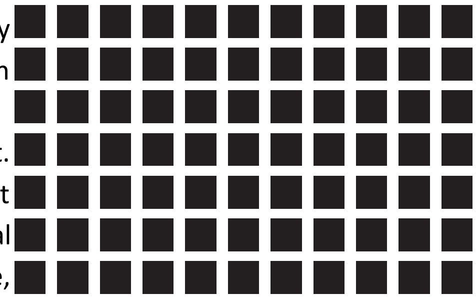
iii. The Knowledge Discovery Pipeline

The knowledge discovery (also called data mining) field has its own pipeline. As with the graphics and visualization pipelines we start with data in this case we process it with the goal of generating a model, rather than some graphics display. The visualization pipeline can be overlaid on this knowledge discovery (KD) pipeline. If we were to look at a pipeline for typical statistical analysis procedures, we would find the same process structure:

- **Data** In the KD pipeline there is more focus on data, as the graphics and visualization processes often assume that the data is already structured to facilitate its display.
- **Data integration** cleaning, warehousing and selection. These involve identifying the various data sets that will be potentially analyzed. Again, the user may participate in this step. This can involve filtering, sampling, subsetting, aggregating, and other techniques that help curate and manage the data for the data mining step.
- **Data mining** The heart of the KD pipeline is algorithmically analyzing the data to produce a model.
- **Pattern evaluation**. The resulting model or models must be evaluated to determine their robustness, stability, precision, and accuracy.
- **Rendering or visualization** The specific results must be presented to the user. It does not matter whether we think of this as part of the graphics or visualization pipelines; the fact is that a user will eventually need to see the results of the process. Model visualization is an exciting research area that will be discussed later.



iv. The Role of Perception In all visualizations a critical aspect related to the user is the abilities and limitations of the human visual system. If the goal of visualization is to accurately convey information with pictures, it is essential that perceptual abilities be considered. There are, of course, no moving black dots at the intersections of the white lines, but clearly such a presentation of data would create instabilities. It thus makes little sense to map a variable to a graphical attribute that humans have limited ability to control or quantify, if the goal is to communicate a numeric value with accuracy.



Users interact with visualizations based upon what they see and interpret. Understanding how we see should help us produce better displays, or at least avoid producing very poor ones. About half of the human brain deals with visual input and much of the processing is parallel and effectively continuous. Texture, color, and motion are examples of primitive attributes that we perceive.

The pipelines briefly discussed earlier, the human perceptual system receives input data and processes it in various ways. The first process is pre attentive processing, a fast, high-performance system that quickly identifies differences in, for example, color or texture. There are other features that the visual system deals with, such as line orientation, length, width, size of an object, curvature, grouping and motion.

Understanding visual perception leads to certain guidelines. For example the Gestalt School of Psychology started in 1912 attempted to define a set of laws by which we perceive patterns. These laws included rules about proximity similarity continuity closure symmetry foreground and background and size.

1.1.3 Pseudo code Conventions: The text we include pseudocode wherever possible. In our pseudocode we aim to convey the essence of the algorithms at hand, while leaving out details required for user interaction, graphics nuances, and data management. We therefore assume that the following global variables and functions exist in the environment of the pseudocode:

- **data**—The working data table. This data table is assumed to contain only numeric values. In practice, dimensions of the original data table that contain non-numeric values must be somehow converted to numeric values. When visualizing a subset of the entire original data table, the working data table is assumed to be the subset.
- **m**—The number of dimensions (columns) in the working data table. Dimensions are typically iterated over using j as the running dimension index.
- **n**—The number of records (rows) in the working data table. Records are typically iterated over using i as the running record index.
- **Normalize** (record, dimension), Normalize (record dimension min max)—A function that maps the value for the given record and dimension in the working data table to a value between \text{min} and \text{max}, or between zero and one if \text{min} and \text{max} are not specified. The normalization is typically linear and local to a single dimension. However, in practice, code must be structured such that various kinds of normalization could be used (logarithmic or square root, for example) either locally (using the bounds of the current dimension), globally (using the bounds of all dimensions), or local to the active dimensions (using the bounds of the dimensions being displayed). Also, in practice, one must accommodate multiple kinds of normalization within a single visualization. For example a scatterplot may require a linear normalization for the x-axis and a logarithmic normalization for the y-axis.
- **Color** (color)—A function that sets the color state of the graphics environment to the specified color (whose type is assumed to be an integer containing RGB values).
- **MapColor** (record, dimension)—A function that sets the color state of the graphics environment to be the color derived from applying the global color map to the normalized value of the given record and dimension in the working data table.

- **Circle** (radius})—A function that fills a circle centered at the given (x, y)-location, with the given radius, with the color of the graphics environment. The plotting space for all visualizations is the unit square. In practice, this function must map the unit square to a square in pixel coordinates.
- **Polyline** (xs, ys)—A function that draws a polyline (many connected line segments) from the given arrays of x- and y-coordinates.
- **Polygon** (xs, ys)—A function that fills the polygon defined by the given arrays of x- and y-coordinates with the color of the current color state.

For geographic visualizations, the following functions are assumed to exist in the environment:

- **GETLATITUDES**(record), GETLONGITUDES(record) — Functions that retrieve the arrays of latitude and longitude coordinates, respectively, of the geographic polygon associated with the given record. For example, these polygons could be outlines of the countries of the world.
- **PROJECTLATITUDES**(lats, scale), PROJECTLONGITUDES(longs, scale) — Functions that project arrays of latitude values to arrays of y values, and arrays of longitude values to arrays of x values, respectively.

For graph and 3D surface data sets, the following is provided:

- **GETCONNECTIONS**(record) — A function that retrieves an array of record indices to which the given record is connected.

1.1.4 The Scatter plot: The scatterplot is one of the earliest and most widely used visualizations developed. It is based on the Cartesian coordinate system. The following pseudocode renders a scatterplot of circles. Records are represented in the scatterplot as circles of varying location, color, and size. The x- and y-axes represent data from dimension numbers xDim and yDim, respectively. The color of the circles is derived from dimension number cDim. The radius of the circles is derived from dimension number rDim, as well as from the upper and lower bounds for the radius, rMin and rMax.

```
SCATTERPLOT(X Dim, yDim, CDim, rDim, rMin, RMax)
for each record i triangleright For each record,
    do x ← NORMALIZE}(i, xDim) triangleright derive the location,
       y ←NORMALIZE}(i, yDim)
       r ←NORMALIZE}(i, rDim, rMin, rMax) triangleright radius,
       MAPCOLOR}(i, cDim) triangleright and color, then
       CIRCLE}(x, y, r) triangleright draw the record as a circle.
```

Vehicle Name	Sedan	Sports	SUV	Wagon	Minivan	Pickup	AWD	RWD	Price
Acura 3.5 RL 4dr	1	0	0	0	0	0	0	0	43755
Acura MDX	0	0	1	0	0	0	1	0	36945
Suzuki XL-7 EX	0	0	1	0	0	0	0	0	23699

which consists of detailed specifications for 428 vehicles. The variables included are dealer and retail price, weight, size, horsepower, fuel efficiency for city and highway, and more. Although it looks like there are 19 variables for each vehicle, columns 2-7 effectively identify the type of car, and columns 8 and 9 the wheel-drive type. Some records have missing entries. The column variables are as follows:

1. Vehicle name (text 1–45 characters);
2. small, sporty, compact or large sedan (1=yes, 0=no);
3. sports car? (1=yes, 0=no);
4. sport utility vehicle? (1=yes, 0=no);
5. wagon? (1=yes, 0=no);
6. minivan? (1=yes, 0=no);
7. pickup? (1=yes, 0=no);
8. all-wheel drive? (1=yes, 0=no);
9. rear-wheel drive? (1=yes, 0=no);
10. suggested retail price, what the manufacturer thinks the vehicle is worth, including adequate profit for the automaker and the dealer (U.S. dollars);

11. dealer cost (or "invoice price"), what the dealership pays the manufacturer (U.S. dollars);
12. engine size (liters);
13. number of cylinders (=-1 if rotary engine);

1.2 Data Foundations: visualization starts with the data that is to be displayed, a first step in addressing the design of visualizations is to examine the characteristics of the data. Data comes from many sources: it can be gathered from sensors or surveys, or it can be generated by simulations and computations. Data can be raw (untreated), or it can be derived from raw data via some process, such as smoothing, noise removal, scaling, or interpolation. It also can have a wide range of characteristics and structures.

A typical data set used in visualization consists of a list of n records, {r1, r2,.., rn}. Each record ri consists of m (one or more) observations or variables, (v1, v2,., vm). An observation may be a single number/symbol/string or a more complex structure (discussed in more detail later in this chapter). A variable may be classified as either independent or dependent. An independent variable ivi is one whose value is not controlled or affected by another variable, such as the time variable in a time-series data set. A dependent variable dvj is one whose value is affected by a variation in one or more associated independent variables. Temperature for a region would be considered a dependent variable, as its value is affected by variables such as date, time, or location.

$$ri = (iv1, iv2,\ldots ivmi, dv1, dv2,\ldots dvmd)$$

where mi is the number of independent variables and md is the number of dependent variables. With this notation we have m = mi + md.

1.2.1 Types of data: its simplest form each observation or variable of a data record represents a single piece of information. We can categorize this information as being ordinal (numeric) or nominal (nonnumeric).

Subcategories of each can be readily defined.

i. **Ordinal** the data take on numeric values:

- binary—assuming only values of 0 and 1;
- discrete—taking on only integer values or from a specific subset (e.g., (2, 4, 6))
- continuous—representing real values (e.g., in the interval [0, 5]).

ii. Nominal the data take on nonnumeric values:

- categorical—a value selected from a finite (often short) list of possibilities (e.g., red, blue, green);
- ranked—a categorical variable that has an implied ordering (e.g., small, medium, large);
- Arbitrary—a variable with a potentially infinite range of values with no implied ordering (e.g., addresses).

Another method of categorizing variables is by using the mathematical concept of scale.

iii. Scale Three attributes that define a variable's measure are as follows:

- Ordering relation with which the data can be ordered in some fashion. By definition ranked nominal variables and all ordinal variables exhibit this relation.
- Distance metric with which the distances can be computed between different records. This measure is clearly present in all ordinal variables but is generally not found in nominal variables.
- Existence of absolute zero in which variables may have a fixed lowest value. This is useful for differentiating types of ordinal variables. A variable such as weight possesses an absolute zero while bank balance does not.

Scale is an important attribute to examine when designing appropriate visualizations because each graphical attribute that we can control has a scale associated with it. Ideally, the scale of a data variable should be compatible with the scale of the graphical entity or attribute to which it is mapped, though it is somewhat dependent on the task to be performed with the visualization.

The type of data also determines the operations that can be applied to the data. While the equality and inequality operators (=, <>) can be applied to any type of data, comparison operators (<, >, <=, >=) can only be applied to ranked nominal and ordinal data types, and mathematical operators and distance computations (+, -, *, /) can only be applied to ordinal data types.

1.2.2 Structure within and between Records: Data sets have structure, both in terms of the means of representation (syntax), and the types of interrelationships within a given record and between records (semantics).

i. Scalars, Vectors, and Tensors

An individual number in a data record is often referred to as a scalar. Scalar values, such as the cost of an item or the age of an individual, are often the focus for analysis and visualization. Multiple variables within a single record can represent a composite data item. For example, a point in a two-dimensional flow field might be represented by a pair of values, such as a displacement in x and y. This pair and any such composition is referred to as a vector. Scalars and vectors are simple variants on a more general structure known as a tensor. A tensor is defined by its rank and by the dimensionality of the space within which it is defined. It is generally represented as an array or matrix.

ii. Geometry and Grids

Geometric structure can commonly be found in data sets especially those from scientific and engineering domains. The simplest method of incorporating geometric structure in a data set is to have explicit coordinates for each data record. A data set of temperature readings from across the country might include the longitude and latitude associated with the sensors as well as the sensor values. In modeling of 3D objects the geometry constitutes the majority of the data with coordinates given for each vertex.

Sometimes the geometric structure is implied. When this is the case it is assumed that some form of grid exists and the data set is structured such that successive data records are located at successive locations on the grid. For example if one had a data set giving elevation at uniform spacing across a surface it would be unnecessary to include the coordinates for each record it would be sufficient to indicate a starting location orientation and the step size horizontally and vertically.

Non-uniform or irregular geometry is also common. For example in simulating the flow of wind around an airplane wing it is important to have data computed densely at locations close to the surface while data for locations far from the surface can be computed much more sparsely. For irregular geometry it is of course essential that the coordinates are explicitly specified in the data file. As the geometry changes from a uniform to a non-uniform or irregular grid the computations for display increase in complexity.

iii. Other Forms of Structure

A timestamp is an important attribute that can be associated with a data record. Time perhaps has the widest range of possible values of all aspects of a data set since we can refer to time with units from picoseconds to millennia. It can also be relative or absolute in terms of its base value. Data sets with temporal attributes can be uniformly spaced such as in the sampling of a continuous phenomenon or non-uniformly spaced as in a business transaction database.

Another important form of structure found within many data sets is that of topology, or how the data records are connected. Connectivity indicates that data records have some relationship to each other. Thus vertices on a surface (geometry) are connected to their neighbors via edges (topology), and relationships between nodes in a hierarchy or graph can be specified by links. Connectivity information is essential to support the processes of resampling and interpolation. This form of structure can either be included explicitly in the data record (e.g., with a fixed or variable length vector identifying the records to which the current record is linked) or as an auxiliary data structure.

The following are examples of various structured data:

- **MRI** (magnetic resonance imagery). Density (scalar) with three spatial attributes 3D grid connectivity
- **CFD** (computational fluid dynamics). Three dimensions for displacement, with one temporal and three spatial attributes, 3D grid connectivity (uniform or non-uniform)
- **Financial** No geometric structure, n possibly independent components, nominal and ordinal, with a temporal attribute
- **CAD** (computer-aided design) Three spatial attributes with edge and polygon connections, and surface properties
- **Remote sensing** Multiple channels, with two or three spatial attributes, one temporal attribute, and grid connectivity

- **Census** Multiple fields of all types, spatial attributes (e.g., addresses), temporal attribute, and connectivity implied by similarities in fields
- **Social Network** Nodes consisting of multiple fields of all types, with various connectivity attributes that could be spatial, temporal or dependent on other attribute such as belonging to the same group or having some common computed values.

1.2.3 Data Preprocessing: In most circumstances it is preferable to view the original raw data. In many domains such as medical imaging the data analyst is often opposed to any sort of data modifications such as filtering or smoothing for fear that important information will be lost or deceptive artifacts will be added. Viewing raw data also often identifies problems in the data set such as missing data or outliers that may be the result of errors in computation or input. Depending on the type of data and the visualization techniques to be applied however some forms of preprocessing might be necessary. Some common methods for preprocessing data are briefly discussed later in this chapter. The interested reader is directed to the many fine textbooks dedicated to these topics for more in-depth coverage.

i. Metadata and Statistics

Information regarding a data set of interest (its metadata) and statistical analysis can provide invaluable guidance in preprocessing the data. Metadata may provide information that can help in its interpretation such as the format of individual fields within the data records. It may also contain the base reference point from which some of the data fields are measured the units used in the measurements, the symbol or number used to indicate a missing value (see below) and the resolution at which measurements were acquired. This information may be important in selecting the appropriate preprocessing operations and in setting their parameters.

The most common statistics about data include the mean and the standard deviation the most common statistical plot is the distribution of data in the form of a histogram.

$$\mu = \frac{1}{n}\sum_{i=0}^{n}(x_i)$$

μ: Population Mean
n: Total Number of Observations
x_i: Individual Data Points

$$\sigma = \sqrt{\sum(x_i - \mu)^2}.$$

ii. Missing Values and Data Cleansing

One of the realities of analyzing and visualizing "real" data sets is that they often are missing some data entries or have erroneous entries. Missing data may be caused by several reasons, including, for example, a malfunctioning sensor, a blank entry on a survey or an omission on the part of the person entering the data. Erroneous data is most often caused by human error and can be difficult to detect.

- **Discard the bad record** data containing a missing or erroneous field is actually one of the most commonly applied since the quality of the remaining data entries in that record may be in question. However this can potentially lead to a significant loss of information especially in data sets containing large numbers of records. In some domains as much as 90% of records have at least one missing or erroneous field
- **Assign a sentinel value** have a designated sentinel value for each variable in the data set that can be assigned when the real value in a record is in question. For example, in a variable that has a range of 0 to 100 one might use a value such as -5 to designate an erroneous or missing entry. The data is visualized the records with problematic data entries will be clearly visible.
- **Assign the average value** simple strategy for dealing with bad or missing data is to replace it with the average value for that variable or dimension. An advantage to using this strategy is that it minimally affects the overall statistics for that variable. The average however may not be a good "guess" for this particular record. Another drawback of using this method is that it may mask or obscure outliers which can be of particular interest.
- **Assign value based on nearest neighbor**. A better approximation for a substitute value is to find the record that has the highest similarity with the record in question based on analyzing the differences in all other variables. The basic idea here is that if record A is missing an entry for variable i and record B is closer than any other record to A without considering variable i then using the value of variable i from record B as a substitute in A is a reasonable assumption.

- **Compute a substitute value** researchers in multivariate statistics have dedicated a significant amount of energy to developing methods for generating values to replace missing or erroneous data. The process, known as imputation, seeks to find values that have high statistical confidence.

iii. Normalization is the process of transforming a data set so that the results satisfy a particular statistical property. A simple example of this is to transform the range of values a particular variable assumes so that all numbers fall within the range of 0.0 to 1.0. Other forms of normalization convert the data such that each dimension has a common mean and standard deviation. Normalization is a useful operation since it allows us to compare seemingly unrelated variables.
For example if dmin and dmax are the minimum and maximum values for a particular data variable, we can normalize the values to the range of 0.0 to 1.0 using the formula

$$\text{dnormalized}=(\text{doriginal}-\text{dmin})/(\text{dmax}-\text{dmin}).$$

To ease interpretation, we sometimes choose the scaling and offset factors to coincide with intuitive maximum and minimum values, rather than the ones the data possess.

$$d_{\text{sqrt-normalized}} = \frac{\left(\sqrt{d_{\text{original}}} - \sqrt{d_{\min}}\right)}{\left(\sqrt{d_{\max}} - \sqrt{d_{\min}}\right)} \qquad d_{\text{log-normalized}} = \frac{\left(\log d_{\text{original}} - \log d_{\min}\right)}{\left(\log d_{\max} - \log d_{\min}\right)}$$

iv. Segmentation the data can be separated into contiguous regions, where each region corresponds to a particular classification of the data. For example, an MRI data set might originally have 256 possible values for each data point, and then be segmented into specific categories such as bone, muscle, fat, and skin. Simple segmentation can be performed by just mapping disjoint ranges of the data values to specific categories. The assignment of values to a category is ambiguous.

```
similarThresh = similarity measure indicating two regions have
                similar characteristics
homogeneousThresh = uniformity measure indicating a region is
                too nonhomogeneous
do {
    changeCount = 0
    for each region
        compare region with neighboring regions to find most
            similar
```

V. Sampling and Subsetting it is necessary to transform a data set with one spatial resolution into another data set with a different spatial resolution. For example we might have an image we would like to shrink or expand or we might have only a small sampling of data points and wish to fill in values for locations between our samples

Vi. Dimension Reduction of the data exceeds the capabilities of the visualization technique, it is necessary to investigate ways to reduce the data dimensionality while at the same time preserving, as much as possible the information contained within. The user to select the dimensions deemed most important or via computational techniques such as principal component analysis (PCA) multidimensional scaling (MDS) Kohonen self-organizing maps (SOMs) and Local Linear Embedding (LLE).

vii. Mapping Nominal Dimensions to Numbers In many domains one or more of the data dimensions consist of nominal values. We may have several alternative strategies for handling these dimensions within our visualizations depending on how many nominal dimensions there are, how many distinct values each variable can take on, and whether an ordering or distance relation is available or can be derived.

Viii. Smoothing and Filtering A common process in signal processing is to smooth the data values, to reduce noise and to blur sharp discontinuities. A typical way to perform this task is through a process known as convolution, which for our purposes can be viewed as a weighted averaging of neighbors surrounding a data point.

1.2.4 Data Sets: The data being displayed, in general, the effectiveness of visualization is enhanced by the user having some context for interpreting what is being shown.

djia-100.xls. A univariate nonspatial data set consisting of 100+ years of daily Dow Jones Industrial Averages.

- Source—analyzeindices.com
- Format—Excel spreadsheet. After the header, each entry is of the form YYMMDD followed by the closing value.
- Code—file can be viewed with Excel.

colorado_elev.vit. A two-dimensional, uniform grid, scalar field representing the elevation of a square region of Colorado.

- Source—included with the distribution of OpenDX (opendx.org).
- Format—binary file with a 268-byte header followed by a 400 × 400 array of 1-byte elevations.
- Code—file can be rendered with TopoSurface, a Processing program included in Appendix C and on the book's web site.

uvw.dat. A three-dimensional uniform grid vector field representing a simulated flow field. The data shows one frame of the unsteady velocity field in a turbulent channel flow, computed by a finite volume method. The streamwise velocity (u) is much larger than the secondary velocities in the transverse direction (v and w).

- Source—Data courtesy of Drs. Jiacai Lu and Gretar Tryggvason ME Department Worcester Polytechnic Institute (me.wpi.edu).
- Format—plain text. After the header, each entry is a set of 6 float values, 3 for position, 3 for displacement. There is roughly a 20:1:1 ratio between the 3 displacements.
- Code—file can be rendered with FlowSlicer a Processing program, and FlowView, a Java program (also need Voxel.java) included in Appendix C and on the book's web site.

city_temp.xls. A two-dimensional nonuniform geo-spatial scalar data set containing the average January temperature for 56 U.S. cities.

- Source_—Peixoto, J.L. (1990), "A Property of Well-Formulated Polynomial Regression Models." American Statistician, 44, 26–30.
- Format—Excel spreadsheet. After a header, each entry is a city name and state, followed by three numbers: average January temperature, latitude, and longitude.
- Code—file can be viewed with Excel.

CThead.zip. A three-dimensional uniform grid, scalar field consisting of a 113-slice MRI data set from a CT scan of a human head.

- Source—Data taken on the General Electric CT Scanner and provided courtesy of North Carolina Memorial Hospital. From graphics.stanford.edu.
- Format—Each slice is stored in a separate binary file (no header). Slices are stored as a 256 × 256 array with dimensions of z=113, y=256, x=256 in z,y,x order. Format is 16-bit integers—two consecutive bytes make up one binary integer (Mac byte ordering). x:y:z shape of each voxel is 1:1:2.
- Code—file can be rendered with VolumeSlicer a Java program included in Appendix C and on the book's web site.

cars.xls, detroit.xls, cereal.xls. Several multivariate, nonspatial data sets commonly used in multivariate data analysis.

- Source_—lib.stat.cmu.edu
- Format_—Excel spreadsheets, mostly with no headers.
- Code_—files can be viewed with Excel.

Health-related multivariate data sets. Subsets from UNICEF's basic indicators by country and the CDC's obesity by state; several multivariate, spatial data sets used with Geospatial Information Systems.

- Source—OpenIndicators.org
- Format—Excel spreadsheets.
- Code—files can be viewed with Excel with Weave and with many other visualization systems.

VAST Contest and Challenge Data. Several heterogeneous data sets that represent realistic scenarios (though they are synthetic) and have embedded ground truth. These were used in the various IEEE VAST contests and challenges.

- Source_—hcil.cs.umd.edu
- Format_—tables, text files, spreadsheet, images, videos, and others.

U.S. Counties Census Data. A subset of the U.S. census for the 3137 counties in the U.S. (including the District of Columbia) that includes county and state total population broken down by age, race, family relationships (number children, age of children) and household parameters (own, rent).